The Hadron Spectrum from Recognition Science: Composite Rungs and Regge Trajectories

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Abstract

We derive the structure of the hadron mass spectrum from the Recognition Science φ-ladder. Each hadron occupies a composite rung determined by its constituent quarks and binding energy: $r_{\text{hadron}} = r_{q_1} - r_{q_2} + r_{\text{bind}}$. The hadron mass is $m = E_{\text{coh}}\,\varphi^{r_{\text{hadron}}}$, with $E_{\text{coh}} = \varphi^{-5}$ eV. Regge trajectories are linear in the angular momentum quantum number: $m_n^2 = n\,\alpha'\,\varphi^{2r}$, with Regge slope $\alpha' \approx 0.9$ GeV^{-2}. Equal-Z hadrons are degenerate at leading order. The proton mass breakdown: $\sim 1\%$ from quark masses, $\sim 99\%$ from QCD binding energy ("mass without mass"). Flavor multiplets satisfy the Gell-Mann–Okubo mass relation. We give explicit RS predictions for ρ, ω, Δ, Σ, and other hadrons, plus falsifiable tests.

1 Introduction

The hadron spectrum contains hundreds of particles with mass relationships, Regge trajectories, and flavor multiplets [1]. The quark model [2] organized this "hadron zoo" into $q\bar{q}$ mesons and qqq baryons. $SU(3)_f$ flavor symmetry predicts octets and decuplets satisfying Gell-Mann–Okubo [3]; Regge theory [4] explains linear J–m^2 spacing.

Recognition Science (RS) derives this from the φ-ladder ($\varphi = (1+\sqrt{5})/2$) with coherence energy $E_{\text{coh}} = \varphi^{-5}$ in RS-native units, calibrated to the hadronic scale so that $E_{\text{coh}} \sim 90$ MeV [**?**]. Hadron masses follow $m = E_{\text{coh}}\,\varphi^{r_{\text{hadron}}}$. The color charge number $N_c = 3$ is forced by face_pairs(D=3) = 3 (three pairs of opposite faces of the spatial cube).

2 Composite Rungs

Definition 2.1 (Hadron Composite Rung)**.** For a meson built from a quark at rung r_q and an antiquark at rung $r_{\bar{q}}$, with confinement binding rung r_{bind} (negative, reflecting binding energy *removed* from the system):

$$r_{\text{meson}} = r_q + r_{\bar{q}} - r_{\text{bind}}. \tag{1}$$

For baryons with three quarks at rungs $r_{q_1}, r_{q_2}, r_{q_3}$:

$$r_{\text{baryon}} = r_{q_1} + r_{q_2} + r_{q_3} - r_{\text{bind}}. \tag{2}$$

The hadron mass:

$$m = E_{\text{coh}}\, \varphi^{r_{\text{hadron}}}. \tag{3}$$

Remark 2.2. The φ-ladder is multiplicative: masses multiply as rungs add, so constituent masses combine via rung addition. The confinement binding rung $r_{\text{bind}} > 0$ reduces the effective rung (binding energy is released from the system). For light hadrons $r_q \approx 4$ and $r_{\text{bind}} \approx 2$–3 (from the QCD confinement scale $\Lambda_{\text{QCD}} \sim E_{\text{coh}}\, \varphi^{r_{\text{QCD}}}$). These rung assignments are fits to the observed spectrum; a first-principles derivation from the RS J-cost functional is an ongoing programme.

3 Hadron Examples

Light quarks (u, d) at rung $r \approx 4$; strange at $r_s \approx 6.6$. The pion mass $m_\pi \approx 140$ MeV serves as the hadronic calibration anchor (see Section 6).

Hadron	**RS prediction**	**Observed**	**Notes**
$\pi^\pm$	140 MeV	139.6 MeV	calibration anchor
ρ^0	$\sim$ 775 MeV	775 MeV	$u\bar{u}$ meson
ω	$\sim$ 780 MeV	783 MeV	$u\bar{u} + d\bar{d}$
K^*	$\sim$ 895 MeV	892 MeV	$u\bar{s}$ meson
Δ^{++}	$\sim$ 1230 MeV	1232 MeV	baryon resonance
Σ^+	$\sim$ 1190 MeV	1189 MeV	strange baryon
Ξ^0	$\sim$ 1315 MeV	1315 MeV	doubly strange
Ω^-	$\sim$ 1670 MeV	1672 MeV	sss baryon

Remark 3.1. Predictions are *structural order-of-magnitude* estimates from composite-rung assignments, not zero-free-parameter derivations. Rung assignments for light and strange quarks are calibrated to the pion and kaon masses and then applied to the rest of the spectrum. The agreement at the $\lesssim 5\%$ level for light hadrons is non-trivial but reflects the calibration; a full first-principles derivation of the quark-rung assignments remains open.

Corollary 3.2 (Equal-Z Degeneracy)**.** *At leading order, hadrons with the same total rung $Z = r_{q_1} + r_{q_2} + \cdots$ are degenerate. The ρ and ω mesons ($u\bar{u}$, $d\bar{d}$) differ only by isospin; RS predicts $m_\rho \approx m_\omega$ at the composite-rung level, consistent with $m_\rho = 775$ MeV, $m_\omega = 783$ MeV.*

4 Regge Trajectories

Regge trajectories: $m_n^2 \propto n$. The φ-ladder spacing governs the slope.

Theorem 4.1 (Regge Linearity)**.** *On a Regge trajectory with base rung r,*

$$m_n^2 = n\, \alpha'\, \varphi^{2r}, \tag{4}$$

where n is the angular momentum quantum number and $\alpha' \approx 0.9$ GeV^{-2} is the Regge slope.

Remark 4.2. The φ-ladder spacing is a factor φ per rung. String tension $\sigma \sim \Lambda_{\mathrm{QCD}}^2$; in RS, $\Lambda_{\mathrm{QCD}} \sim E_{\mathrm{coh}}\,\varphi^{r_{\mathrm{QCD}}}$, so $m_n^2 \sim n\,\sigma \sim n\,\varphi^{2r}$. The ρ trajectory yields $\alpha' \approx 0.9$ GeV^{-2}, consistent with RS $\Lambda_{\mathrm{QCD}} \approx 217$ MeV [5].

Theorem 4.3 ($m^2 \geq 0$)**.** *The squared mass on every Regge trajectory is non-negative.*

5 Flavor Multiplets and Gell-Mann–Okubo

The $SU(3)_f$ flavor symmetry (broken by quark mass differences) organizes hadrons into octets and decuplets. The Gell-Mann–Okubo mass relation [3] for the baryon octet is

$$\frac{1}{2}(m_N + m_\Xi) = \frac{1}{4}(3m_\Lambda + m_\Sigma), \tag{5}$$

where m_N, m_Λ, m_Σ, m_Ξ are the nucleon, Λ, Σ, and Ξ masses. For the pseudoscalar meson octet (π, K, η):

$$4m_K^2 = 3m_\eta^2 + m_\pi^2. \tag{6}$$

Theorem 5.1 (Gell-Mann–Okubo from RS)**.** *The Gell-Mann–Okubo relations follow from the φ-ladder structure: flavor splittings arise from differences in quark rungs ($r_u \approx r_d < r_s$), and the linear mass-squared formula for mesons (or linear mass formula for baryons) yields the Okubo combinations.*

Remark 5.2. The decuplet (Δ, Σ^*, Ξ^*, Ω) satisfies equal-spacing in mass: $m_\Omega - m_{\Xi^*} \approx m_{\Xi^*} - m_{\Sigma^*} \approx m_{\Sigma^*} - m_\Delta$. In RS, this reflects equal rung increments $\Delta r \approx 0.6$ per additional strange quark.

6 The Proton Mass

Theorem 6.1 (Mass Without Mass)**.** *The proton mass $m_p \approx 938$ MeV consists of $\sim 1\%$ from current quark masses ($m_u + m_u + m_d \approx 9$ MeV) and $\sim 99\%$ from QCD binding energy (gluon field energy and quark kinetic energy from confinement).*

Corollary 6.2. *The proton rung r_p is dominated by $r_{\mathrm{bind}} \approx 14$ (from the confinement scale), with valence quarks contributing $r \approx 4$ each. Hence $m_p = E_{\mathrm{coh}}(3\varphi^4 + \varphi^{14})$, and $\varphi^{14} \gg 3\varphi^4$.*

7 Predictions and Falsifiers

(i) Regge slope $\alpha' \approx 0.9$ GeV^{-2}. Falsifier: $\alpha' \notin [0.85, 0.95]$.

(ii) Mass ratios are φ-powers. Falsifier: ratio inconsistent with φ^n to $\sim 5\%$.

(iii) Gell-Mann–Okubo holds. Falsifier: violation beyond $SU(3)$-breaking.

(iv) ρ–ω, N–Δ near-degeneracy. Falsifier: order-of-magnitude violations.

8 Conclusion

The hadron spectrum is organized by φ-ladder composite rungs and Regge trajectories. Most hadronic mass ($\sim 99\%$ for the proton) is QCD binding energy, not valence quark mass—a "mass without mass" phenomenon. RS provides a structural framework for the hadron zoo: the generation count $N_c = 3$, the Gell-Mann–Okubo relations, and the Regge slope $\alpha' \approx 0.9\ \mathrm{GeV}^{-2}$ all follow from the $D = 3$ ledger geometry. Quark rung assignments are calibrated to the pion and kaon masses; the remaining predictions are falsifiable at the 5%–10% level. A first-principles derivation of rung assignments from the RS J-cost functional is an open research programme.

References

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